cogs 9 | Week 3

Data wrangling

Brendan Tomoschuk

Reminders

Late adds + International students

1. Reading Quiz 1, syllabus quiz open until **FRIDAY WEEK 3**

Reading Quiz 2

Due **this** Wednesday!

Reading Quiz 3

Due **next** Wednesday!

Group Project Guidelines

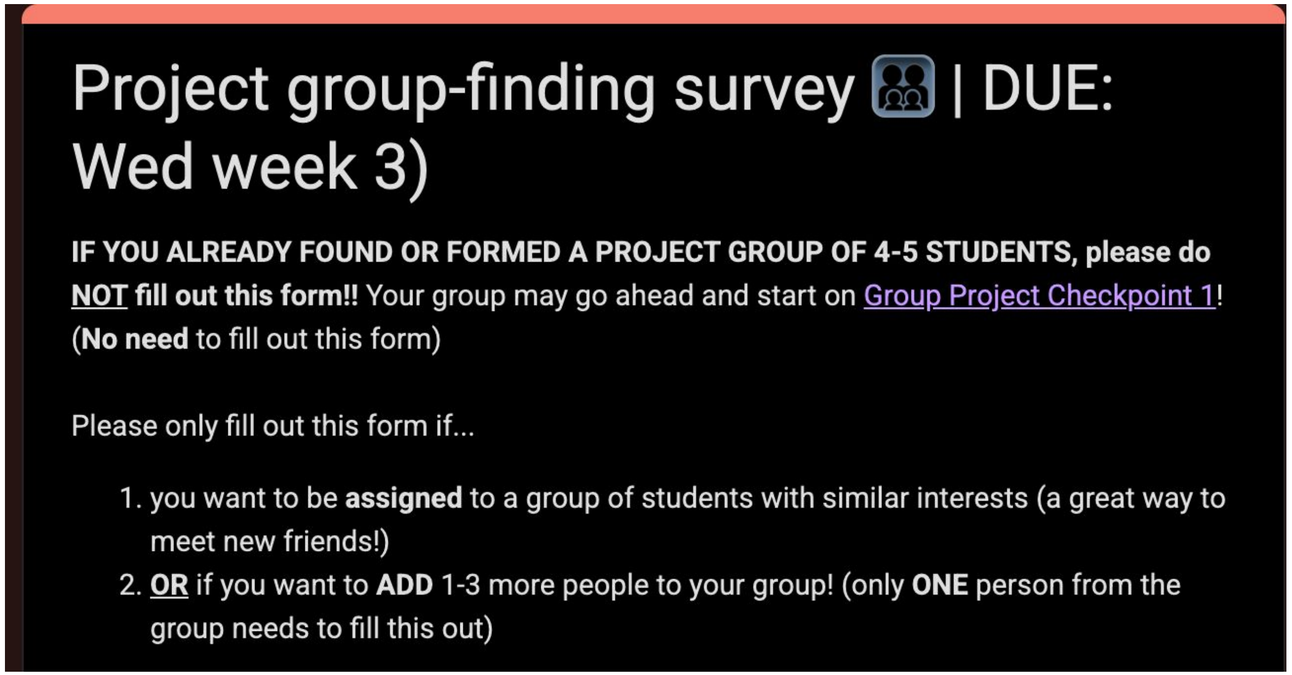
Groups of 4-5 students

1. Form your own group **OR**
2. Be matched with a group based on interests / work styles

Group Project Checkpoint 1 is

DUE Wednesday of Week 5

- If you want to be matched **OR** want 1-3 more people in your group, please fill out the group-finding survey by **Wednesday of week 3!**
  - We will match groups by the **end of that week BY EMAIL** **(contact your group ASAP)**



Midterm 1 — Monday October 19

- **Format:** 35 multiple-choice questions, 50 mins, ***CLOSED-NOTE***
  - ***Questions taken from***
    - ***lectures up to and including “Wrangling data”***
    - ***Readings 1-3***
- **Bring:**
  - A pen or pencil
  - No need to bring a scantron
- **Reminders:**
  - Come early and get a seat to maximize everyone’s time!
  - Students with accommodations will take the exam at the Triton Testing Center. Email me ASAP if you have not sent me your accommodations letter

Guest Lecture — Friday, October 23, 2026

Data Exploration & Startup life

- Participation is worth **1 point**
  - Fill out a Google Form of your takeaways
  - Interactive **Q&A!**
- Dewey data organizes and sells datasets to academic researchers
- Questions examples:
  - Working at at a small company?
  - What kind of data do you work with?



Taylor Butler

Research & Growth Lead

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course code:
WSTAPX

Which of the following is NOT a reason to use APIs instead of web scraping?

(**A**) APIs usually provide data in a semi-structured, ready-to-use format such as JSON

(**B**) APIs give you full control over all the data attributes you want to collect

(**C**) APIs often include official documentation and usage guidelines

(**D**) APIs are generally more stable and less likely to break when a website layout changes

(**E**) I'm not sure / I'm confused

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course code:
WSTAPX

Which of the following is NOT a reason to use APIs instead of web scraping?

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course code:

WSTAPX

You want to analyze daily COVID-19 case counts for every U.S. state over the past 3 years. What is the <u>best way</u> to obtain this data?

(A) Collect it yourself

(B) Steal it

(C) Buy it

(D) Use public / open-source data

(E) I'm not sure / I'm confused

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course code:
WSTAPX

You want to analyze daily COVID-19 case counts for every U.S. state over the past 3 years. What is the best way to obtain this data?

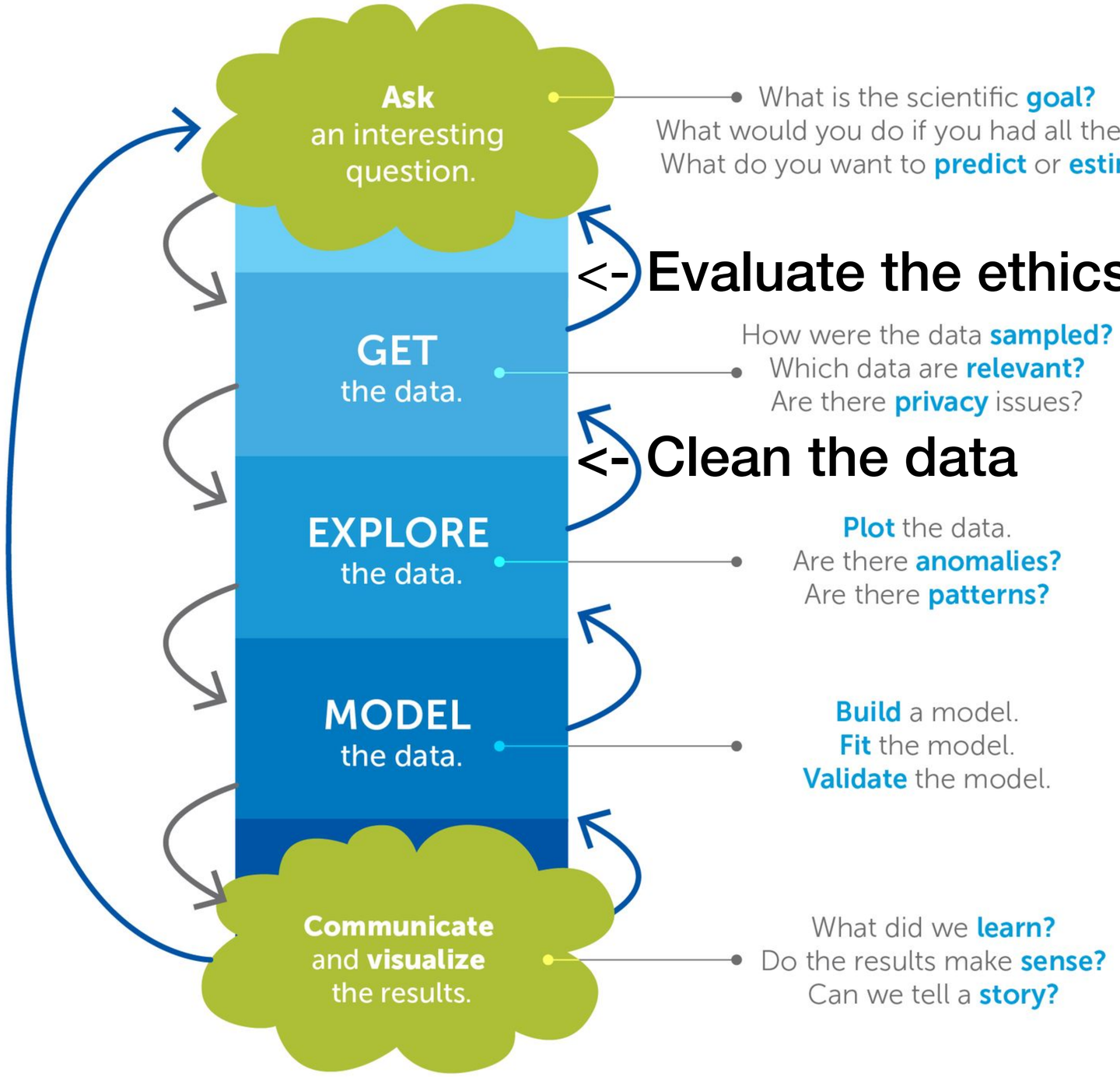
(A) Collect it yourself

(B) Steal it

(C) Buy it

(D) Use public / open-source data

(E) I'm not sure / I'm confused



The data science process

Credit: Joe Blitzen & Hanspeter Pfister (Harvard CS 109)

Learning Goals

1. Describe **best practices** for storing tabular data
2. Explain the **tidy data principles**
3. Identify common data quality issues (missing values, duplicates, inconsistent formats, outliers)
4. **Propose solutions** for wrangling a messy dataset into the tidy data format

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course code:

WSTAPX

What percentage of a data scientist's time is spent preparing data so that it can be used to answer an interesting question (also known as *data wrangling*)?

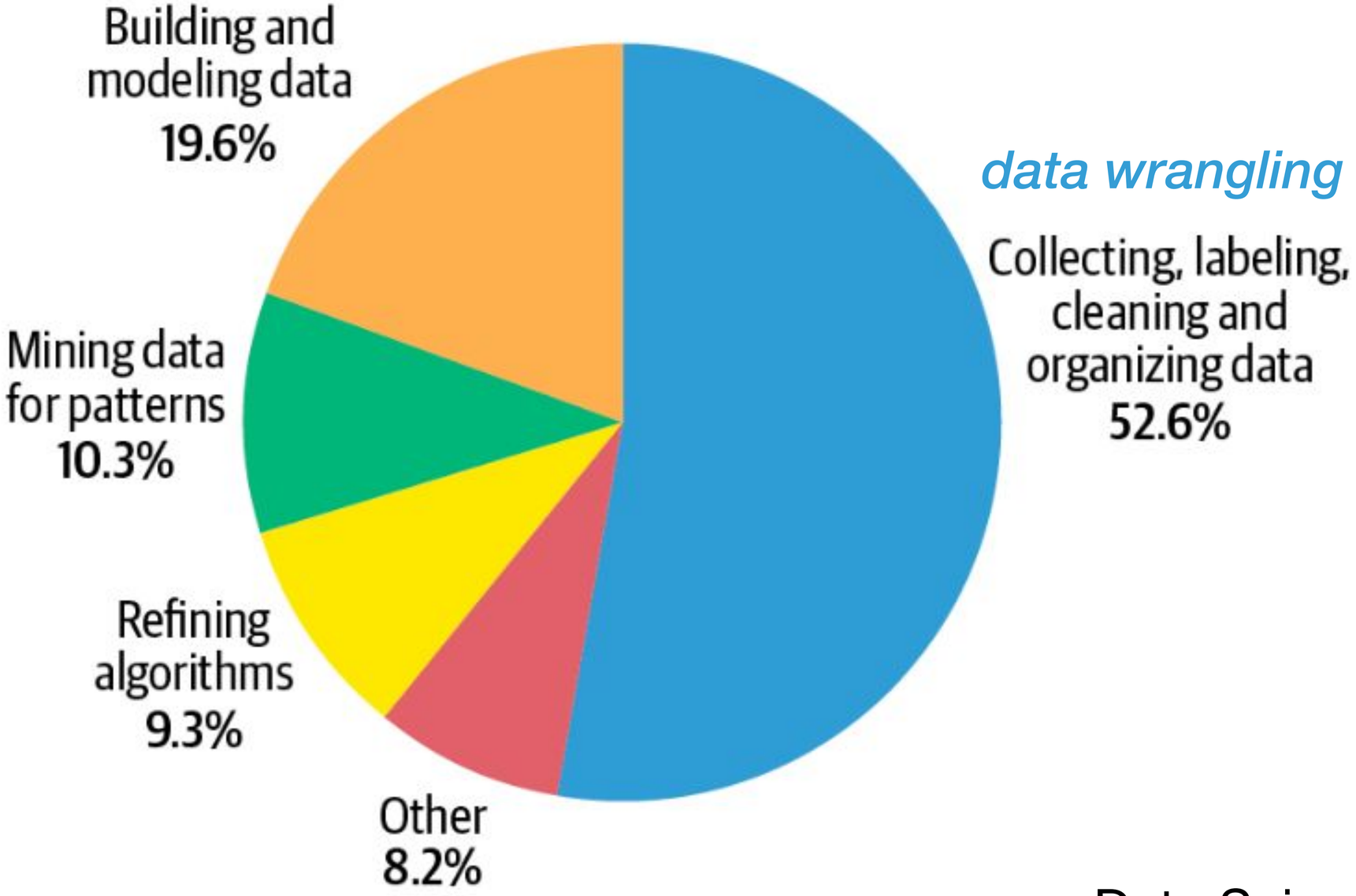
(A) 0-20%

(B) 20-40%

(C) 40-60%

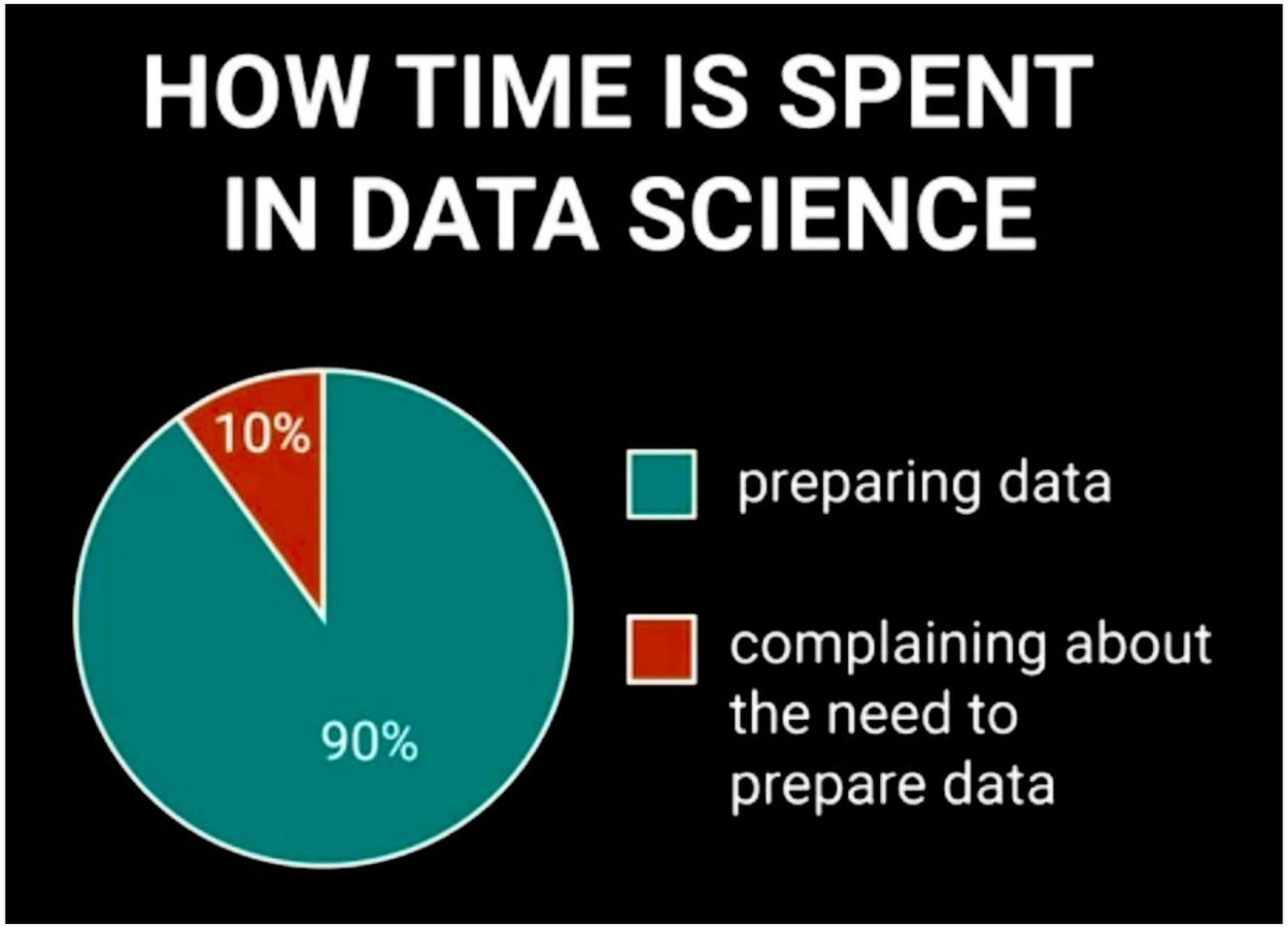
(D) 60-80%

(E) 80-100%



data wrangling

Data Scientist Report


HOW TIME IS SPENT
IN DATA SCIENCE
10%
90%
preparing data
complaining about
the need to
prepare data



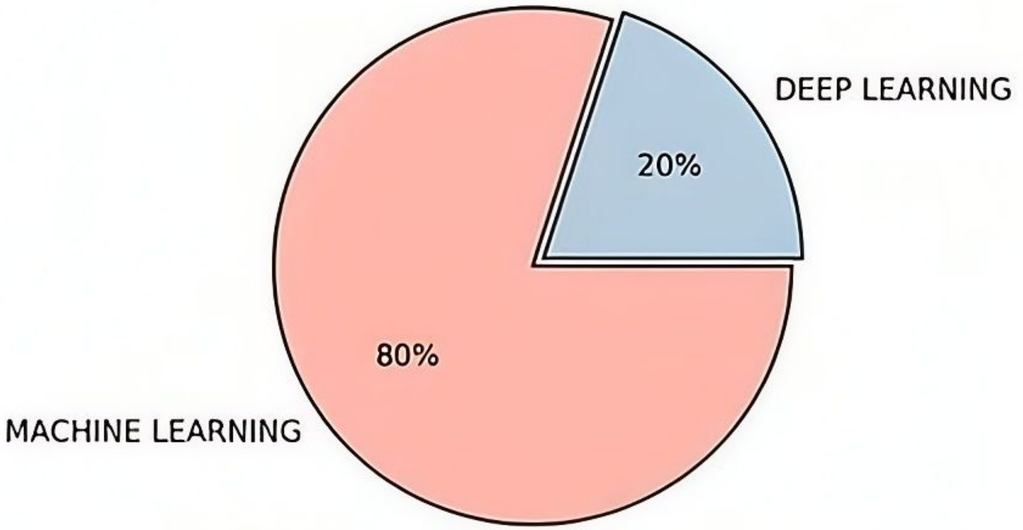


When I started cleaning data

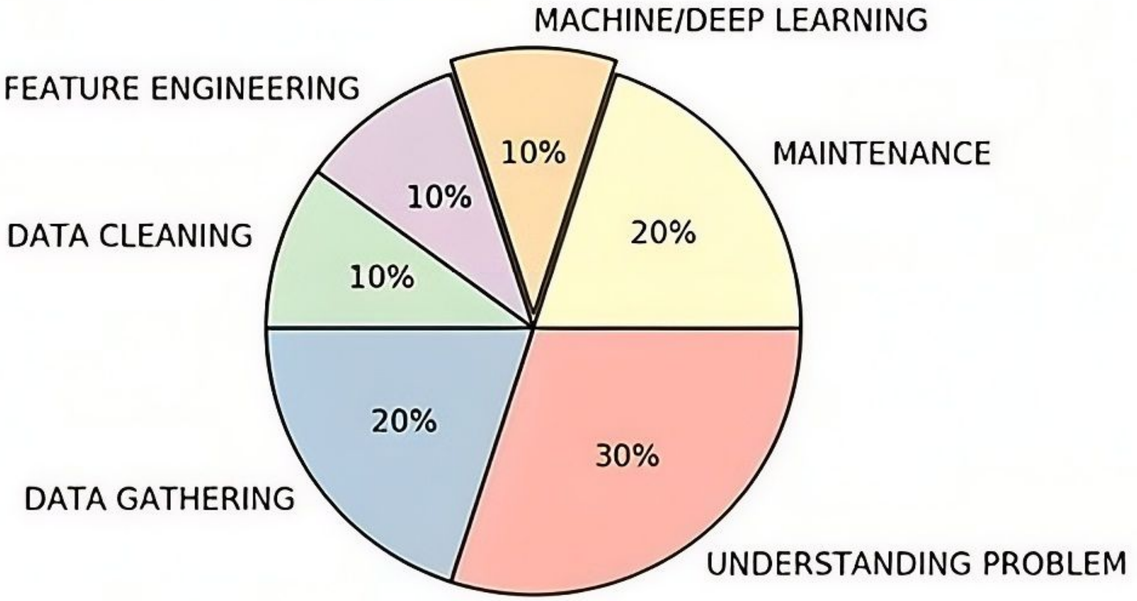
When I finished cleaning data



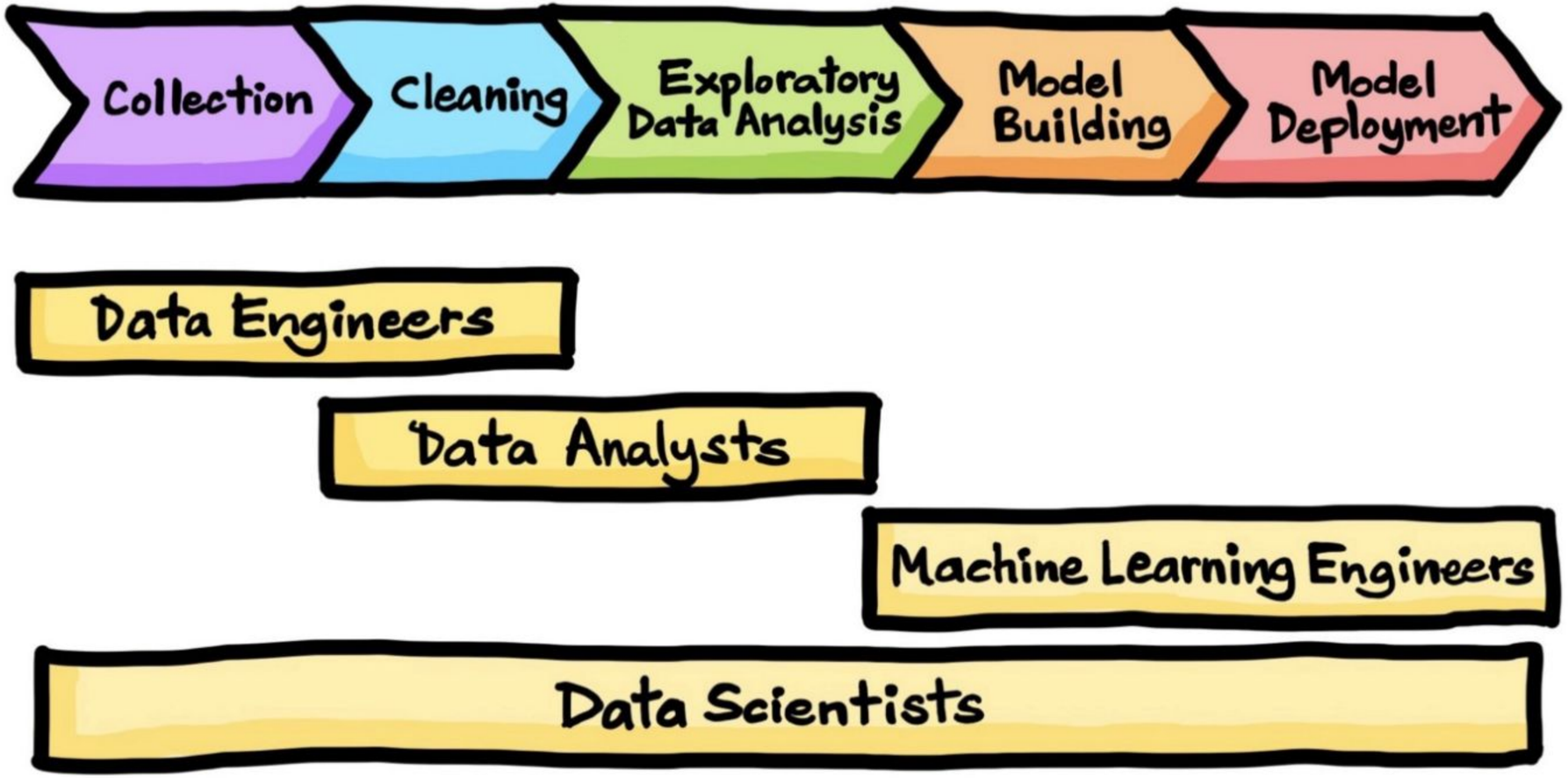
DATA SCIENTIST JOB - EXPECTATION



DATA SCIENTIST JOB - REALITY



Data science careers



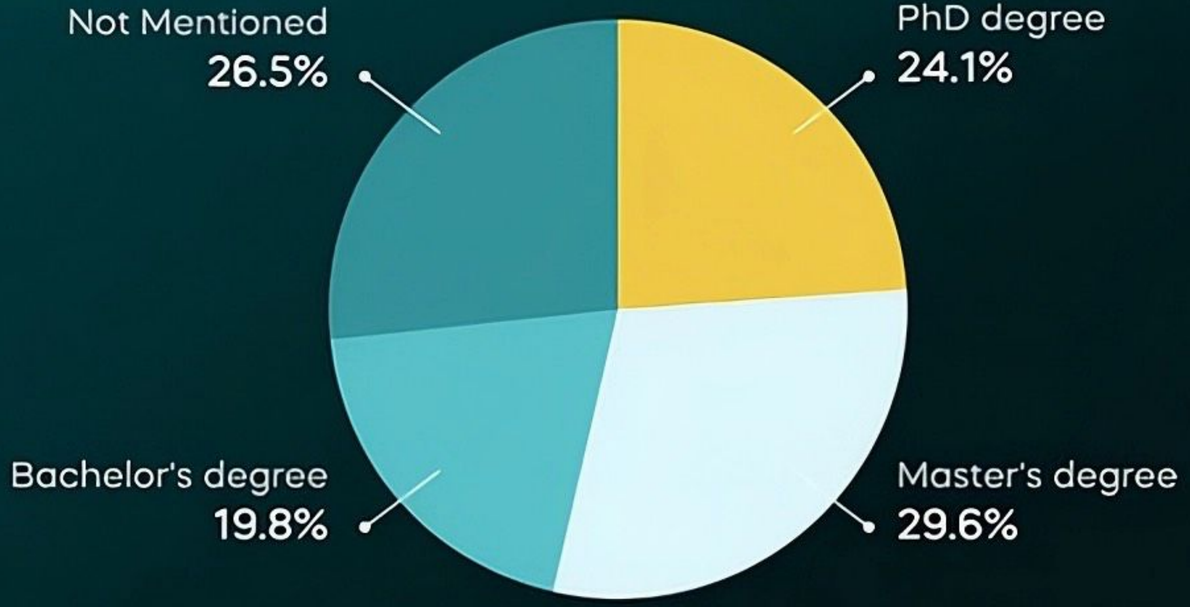
Credit: towardsdatascience.com

365

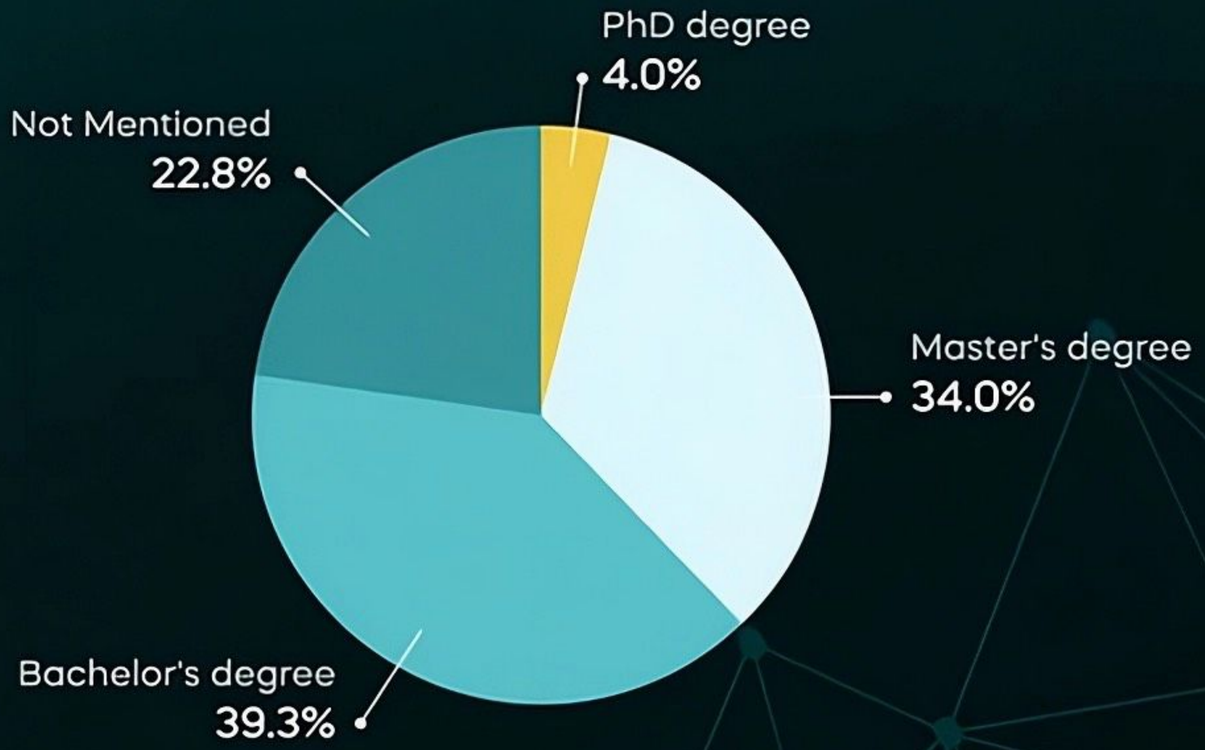
Data Scientist vs Data Engineer

Education Level

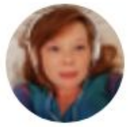
Data Scientist



Data Engineer



Percentage of Mentions in Total Job Postings



Kat Scott 🐁
@kscottz

One of the biggest failures I see in junior ML/CV engineers is a complete lack of interest in building data sets. While it is boring grunt work I think there is so much to be learned in putting together a dataset. It is like half the problem.

11:50 AM · Feb 1, 2019



Kat Scott 🐁
@kscottz

At this point I consider my job to be 95% building tools to scrape, mine, move, annotate, review, and preprocess data sets, and about 5% doing the actual "machine learning".

11:50 AM · Feb 1, 2019

7 29 144 3



ImageNet: The dataset that changed the development of AI



Dr. Fei-Fei Li (李飞飞)

Stanford University

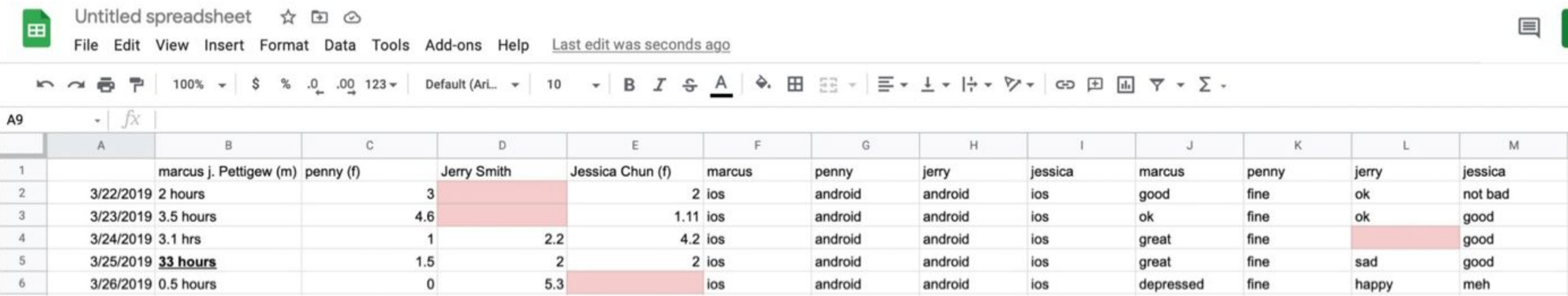
Set up by data scientist Fei-Fei Li in 2006, the ImageNet database now contains more than 14 million annotated images. It has played a key role in advancing computer vision across applications like object recognition, image classification, and object localization.

“Good data scientists understand, in a deep way, that the heavy lifting of cleanup and preparation isn't something that gets in the way of solving the problem: it is the problem.”

—Dhanurjay "DJ" Patil
US Chief Data Scientist, Obama White House

What variables do you see in the dataset?
What's wrong with how this dataset is organized?

(~4 min with your neighbor)



	A	B	C	D	E	F	G	H	I	J	K	L	M
1		marcus j. Pettigew (m)	penny (f)	Jerry Smith	Jessica Chun (f)	marcus	penny	jerry	jessica	marcus	penny	jerry	jessica
2	3/22/2019	2 hours	3		2	ios	android	android	ios	good	fine	ok	not bad
3	3/23/2019	3.5 hours	4.6		1.11	ios	android	android	ios	ok	fine	ok	good
4	3/24/2019	3.1 hrs	1	2.2	4.2	ios	android	android	ios	great	fine		good
5	3/25/2019	**33 hours**	1.5	2	2	ios	android	android	ios	great	fine	sad	good
6	3/26/2019	0.5 hours	0	5.3		ios	android	android	ios	depressed	fine	happy	meh

What variables do you see in the dataset?
What's wrong with how this dataset is organized?

(~4 min with your neighbor)

	A	B	C	D	E
1		marcus j. Pettigew (m)	penny (f)	Jerry Smith	Jessica Chun (f)
2	3/22/2019	2 hours	3		2
3	3/23/2019	3.5 hours	4.6		1.11
4	3/24/2019	3.1 hrs	1	2.2	4.2
5	3/25/2019	**33 hours**	1.5	2	2
6	3/26/2019	0.5 hours	0	5.3	

...

F	G	H	I	J	K	L	M
marcus	penny	jerry	jessica	marcus	penny	jerry	jessica
ios	android	android	ios	good	fine	ok	not bad
ios	android	android	ios	ok	fine	ok	good
ios	android	android	ios	great	fine		good
ios	android	android	ios	great	fine	sad	good
ios	android	android	ios	depressed	fine	happy	meh

Australian Bureau of Statistics

Table junk

1800.0 Australian Marriage Law Postal Survey, 2017

Released on 15 November 2017

Table 5 Participation by Federal Electoral Division(a), Males and Age Gender apartheid

Yeah NA		18-19 years	20-24 years	25-29 years	30-34 years	35-39 years	40-44 years	45-49 years	50-54 years	55-59 years	60-64 years
Lingian(c)	Total participants	292	1,058	1,465	1,653	1,515	1,516	1,710	1,730	1,753	1,574
	Eligible participants	572	2,910	3,789	3,996	3,607	3,506	3,645	3,331	2,960	2,456
Primary keynotes	Participation rate (%)	51.0	36.4	38.7	41.4	42.0	43.2	46.9	51.9	59.2	64.1
				Comma on							
Merged cells											
Solomon	Total participants	442	1,461	2,066	2,357	2,188	2,057	2,224	2,108	2,134	1,772
	Eligible participants	750	2,991	3,994	4,155	3,634	3,398	3,427	3,066	2,931	2,355
	Participation rate (%)	58.9	48.8	51.7	56.7	60.2	60.5	64.9	68.8	72.8	75.2
Northern Territory (Total)	Total participants	734	2,519	3,531	4,010	3,703	3,573	3,934	3,838	3,887	3,346
	Eligible participants	1,322	5,901	7,783	8,151	7,241	6,904	7,072	6,397	5,891	4,811
	Participation rate (%)	55.5	42.7	45.4	49.2	51.1	51.8	55.6	60.0	66.0	69.5
					Summary of data inside data						
Australian Capital Territory Divisions	Covariate as Subheading										
Canberra(d)	Total participants	1,764	4,789	4,817	4,973	4,626	4,453	5,074	4,826	5,169	4,394
	Eligible participants	2,260	6,471	6,448	6,509	5,983	5,805	6,302	5,902	6,044	5,057
	Participation rate (%)	78.1	74.0	74.7	76.4	77.3	76.7	80.5	81.8	85.5	86.9
Fenner(e)	Total participants	1,477	4,687	5,178	5,786	6,025	5,463	5,191	4,208	3,948	3,465
	Eligible participants	1,904	6,354	7,121	7,822	7,960	7,155	6,480	5,206	4,692	3,945
	Participation rate (%)	77.6	73.8	72.7	74.0	75.7	76.4	80.1	80.8	84.1	87.8
				NA Yeah							
Australian Capital Territory (Total)	Total participants	3,241	3,476	3,393	10,759	10,651	3,910	10,265	3,034	3,117	7,659
	Eligible participants	4,164	12,825	13,569	14,331	13,943	12,960	12,782	11,108	10,736	9,002
	Participation rate (%)	77.8	73.9	73.7	75.1	76.4	76.5	80.3	81.3	84.9	87.3
Australia											
Total	Total participants	151,297	438,166	441,658	460,548	462,206	479,360	524,620	517,693	543,449	506,799
	Eligible participants	201,439	635,909	646,916	665,250	656,446	660,841	693,850	659,150	664,720	597,386
	Participation rate (%)	75.1	68.9	68.3	69.2	70.4	72.5	75.6	78.5	81.8	84.8

(a) The Federal Electoral Divisions are current as at 24 August 2017
(b) Includes those whose age is unknown
(c) Includes Christmas Island and the Cocos (Keeling) Islands
(d) Includes Norfolk Island
(e) Includes Jervis Bay

Return of the table junk

MS Excel or Die

data wrangling script

```r
age_counts <- readxl::read_excel(
  path = file_name,
  range = "Table 5!B8:R650",
  col_names = c("measure",
                "18–19 years",
                "20–24 years",
                "25–29 years",
                "30–34 years",
                "35–39 years",
                "40–44 years",
                "45–49 years",
                "50–54 years",
                "55–59 years",
                "60–64 years",
                "65–69 years",
                "70–74 years",
                "75–79 years",
                "80–84 years",
                "85 years and over",
                "Total for Gender")) %>%
  drop_na() %>%
  mutate(area = rep(areas$area, each=3), gender="Male") %>%
  filter(!grepl("Total", area)) %>%
  gather(key = "age", value = "count", -measure, -area, -gender)
}
```

tidy_aus_ssm2.R hosted with ❤️ by GitHub view raw

area	gender	age	State	Area (sq km)	Eligible participants	Participation rate (%)	Total participants	Total Paticipants
Adelaide	Female	18-19 years	SA	76	1341	83.5	1120	1120
Adelaide	Female	20-24 years	SA	76	4620	81.2	3750	3750
Adelaide	Female	25-29 years	SA	76	4897	81.8	4004	4004
Adelaide	Female	30-34 years	SA	76	4784	79.8	3820	3820
Adelaide	Female	35-39 years	SA	76	4319	79	3411	3411
Adelaide	Female	40-44 years	SA	76	4310	80.6	3472	3472
Adelaide	Female	45-49 years	SA	76	4579	81.4	3728	3728
Adelaide	Female	50-54 years	SA	76	4475	84.7	3791	3791
Adelaide	Female	55-59 years	SA	76	4622	87.3	4033	4033
Adelaide	Female	60-64 years	SA	76	4342	89.3	3879	3879
Adelaide	Female	65-69 years	SA	76	3970	90.7	3602	3602
Adelaide	Female	70-74 years	SA	76	3009	90.3	2716	2716
Adelaide	Female	75-79 years	SA	76	2156	88.5	1908	1908
Adelaide	Female	80-84 years	SA	76	1673	85.1	1423	1423

→ *data wrangling turns it into tidy data!*

Structured data

- **Structured data** can be organized into an *n x p* table of values

- *n* = the number of **observations**, or sample size **(rows)**

- *p* = the number of **features or variables (columns)**

- Variables can be quantitative or qualitative

n = 12 observations

p = 4 features OR variables

College dataset

	Private	Apps	Accept	Enroll
Abilene Christian University	Yes	1660	1232	721
Adelphi University	Yes	2186	1924	512
Adrian College	Yes	1428	1097	336
Agnes Scott College	Yes	417	349	137
Alaska Pacific University	Yes	193	146	55
Albertson College	Yes	587	479	158
Albertus Magnus College	Yes	353	340	103
Albion College	Yes	1899	1720	489
Albright College	Yes	1038	839	227
Alderson-Broaddus College	Yes	582	498	172
Alfred University	Yes	1732	1425	472
Allegheny College	Yes	2652	1900	484

qualitative quantitative

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course code:
WSTAPX

How many observations and variables does this dataset have?

person	treatment	result
John Smith	a	—
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

(A) 3 observations, 6 variables

(B) 6 observations, 3 variables

(C) 3 observations, 3 variables

(D) I'm not sure / I'm confused

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course code:
WSTAPX

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Tidy Data

1. Each variable forms a column
2. Each observation forms a row
3. Each type of observational unit forms a table



Dr. Hadley Wickham

RStudio, creator of Tidyverse



Tidy Data

How many variables?

1. Each variable forms a column
2. Each observation forms a row
3. Each type of observational unit forms a table

Variables

- user_id
- name
- age
- sex

user_id	name	age	sex
1	Marcus J.	29	M
2	Aisha Khan	34	F
3	Lee Chen	22	F

Tidy Data

How many observations?

1. Each variable forms a column
2. Each observation forms a row
3. Each type of observational unit forms a table

user_id	name	age	sex
1	Marcus J.	29	M
2	Aisha Khan	34	F
3	Lee Chen	22	F

Tidy Data

1. Each variable forms a column
2. Each observation forms a row
3. **Each type of observational unit forms a table**

One table *per type of data!*

user_id	name	age	sex
1	Marcus J.	29	M
2	Aisha Khan	34	F
3	Lee Chen	22	F

User demographic data

usage_id	user_id	date	device_os	screen_time_min	emotion_rating
1	1	2026-01-15	iOS	42	3
3	2	2026-01-15	Android	58	4
4	3	2026-01-16	Android	15	1

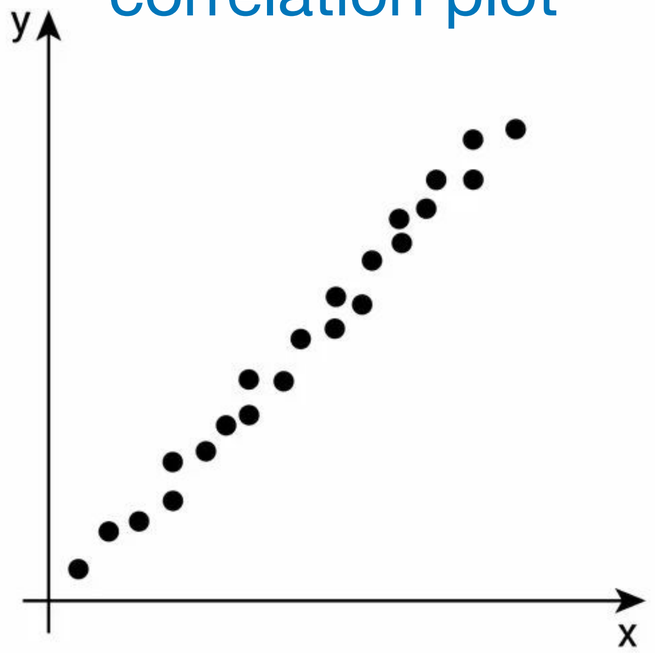
user_id links the two tables

Daily device usage data

Benefits of Tidy Data

1. **Easier to analyze and model:** most stats/ML tools work best when data are in "rows = observations, columns = variables" shape
   - e.g., plotting a correlation between two variables = two columns
2. **Reduces errors:** Clear structure prevents double-counting data or mixing units
3. **Consistent structure makes it reusable and shareable:** Other people (and a future version of you) can easily use the data because the layout is predictable

correlation plot



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course code:
WSTAPX

According to tidy data principles, what is wrong with this table?

	treatmenta	treatmentb
John Smith	—	2
Jane Doe	16	11
Mary Johnson	3	1

(A) The table contains missing values

(B) The treatment labels are not capitalized consistently

(C) The same variable is stored across multiple columns instead of one column

(D) I'm not sure / I'm confused

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WSTAPX

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(D) I'm not sure / I'm confused

Messy data

	treatmenta	treatmentb
John Smith	—	2
Jane Doe	16	11
Mary Johnson	3	1

Tidy data

person	treatment	result
John Smith	a	—
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

Next time

1. Tidy spreadsheets
2. Data dictionaries